Project Title: Deploying and Configuring Azure Firewall with Threat Protection

Author: Precious Iwuozor

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INTRODUCTION

This project demonstrates how to securely deploy and configure an Azure Firewall from scratch. It includes setting up the network infrastructure, configuring NAT, application, and network rules, and enabling threat intelligence protection. The goal is to establish a robust perimeter defence for cloud resources using Microsoft Azure.

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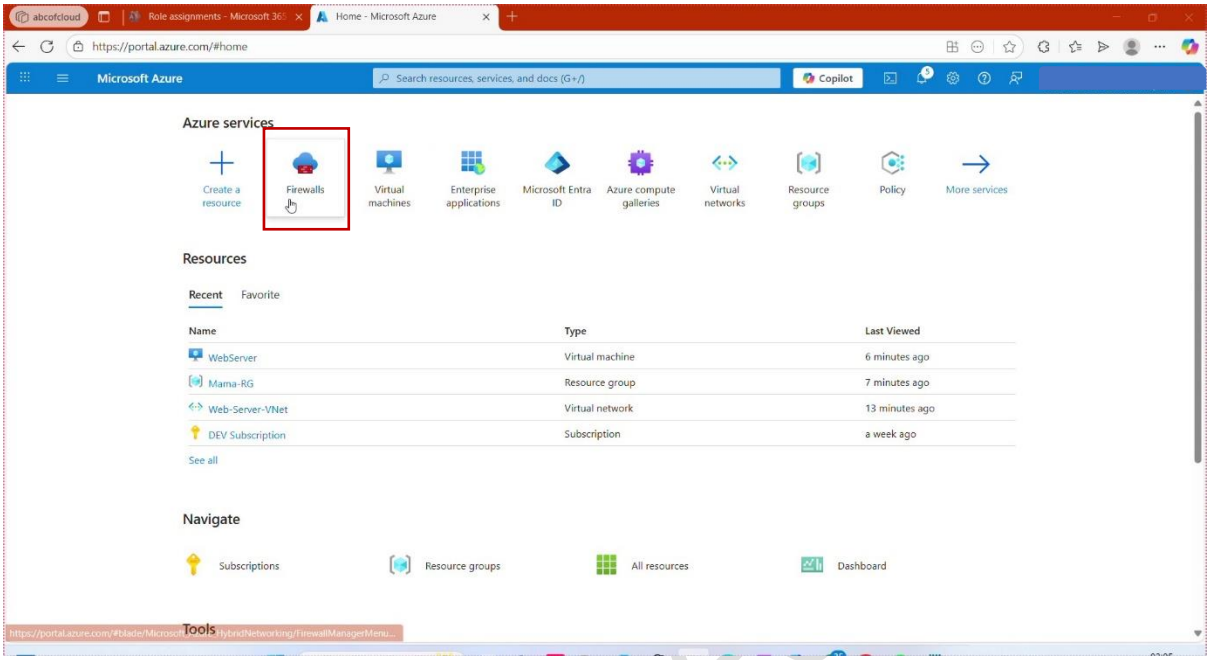
1. Tools & Environment

- Microsoft Azure Portal
- Azure Firewall
- Virtual Network (VNet)
- Subnets (FirewallSubnet, AzureFirewallManagementSubnet)
- Public IP Address (PIP)
- Threat Intelligence Protection

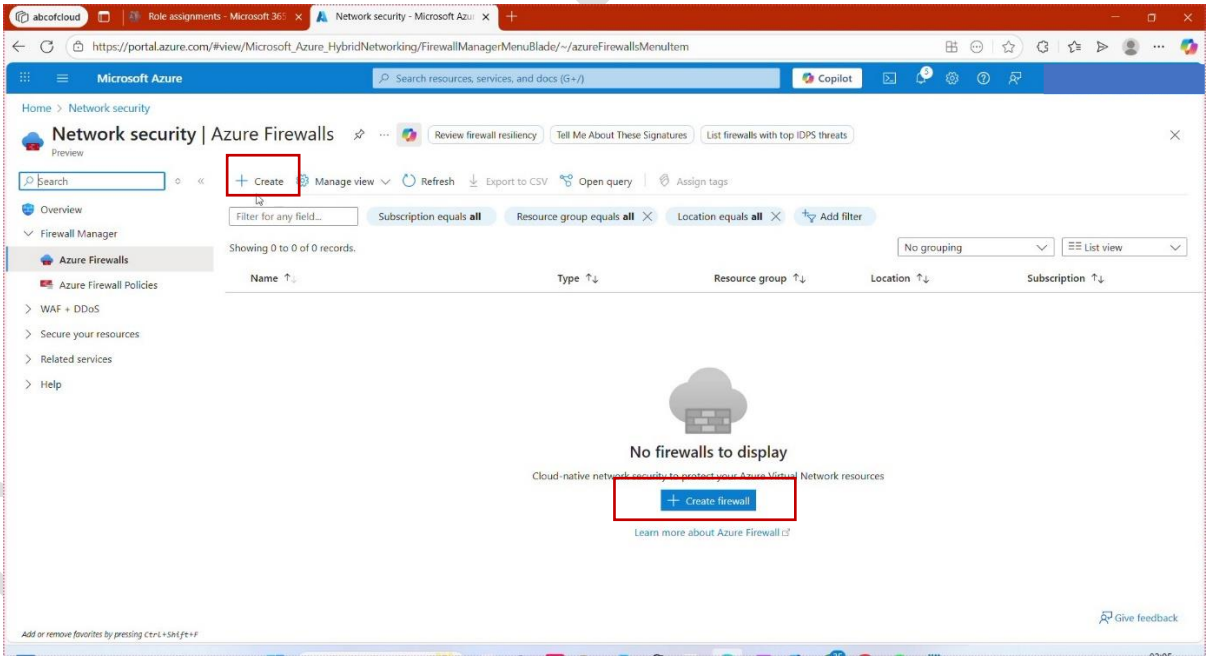
2. Step 1: Create a Resource Group

- Navigate to **Azure Portal** → **Resource Groups**
- Click **Create**
- Name your **resource group**
- Select **region** and click R**eview** + **Create**

1. **On the Azure home page select Firewalls under the Azure services bar.**

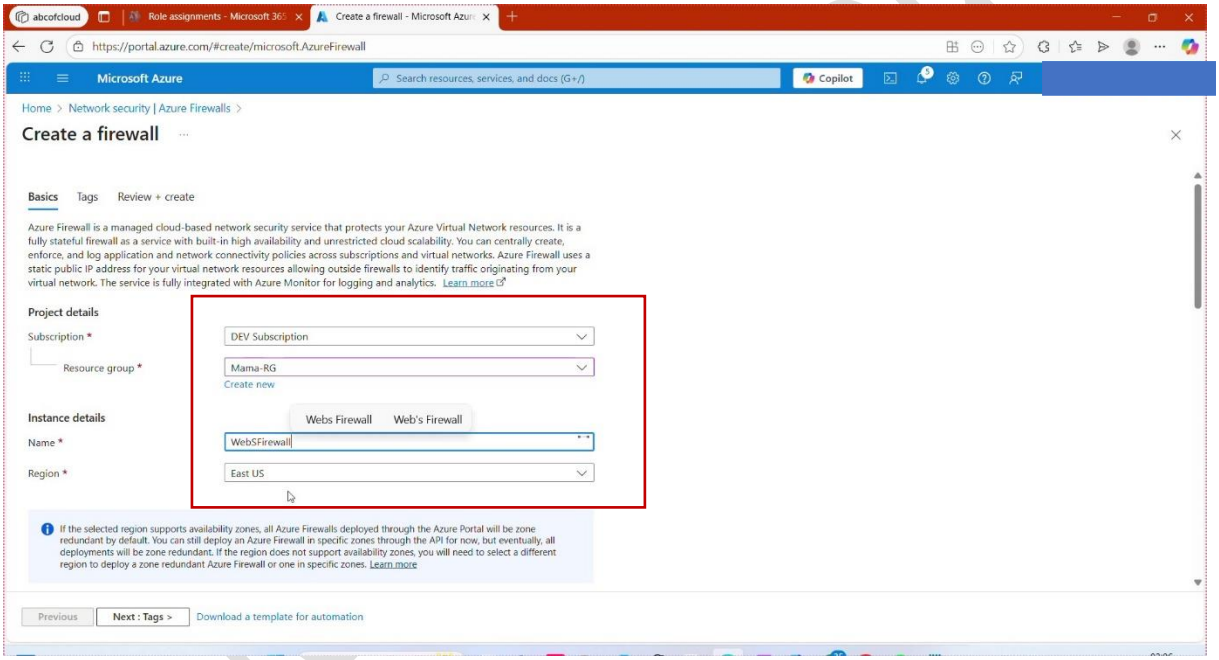


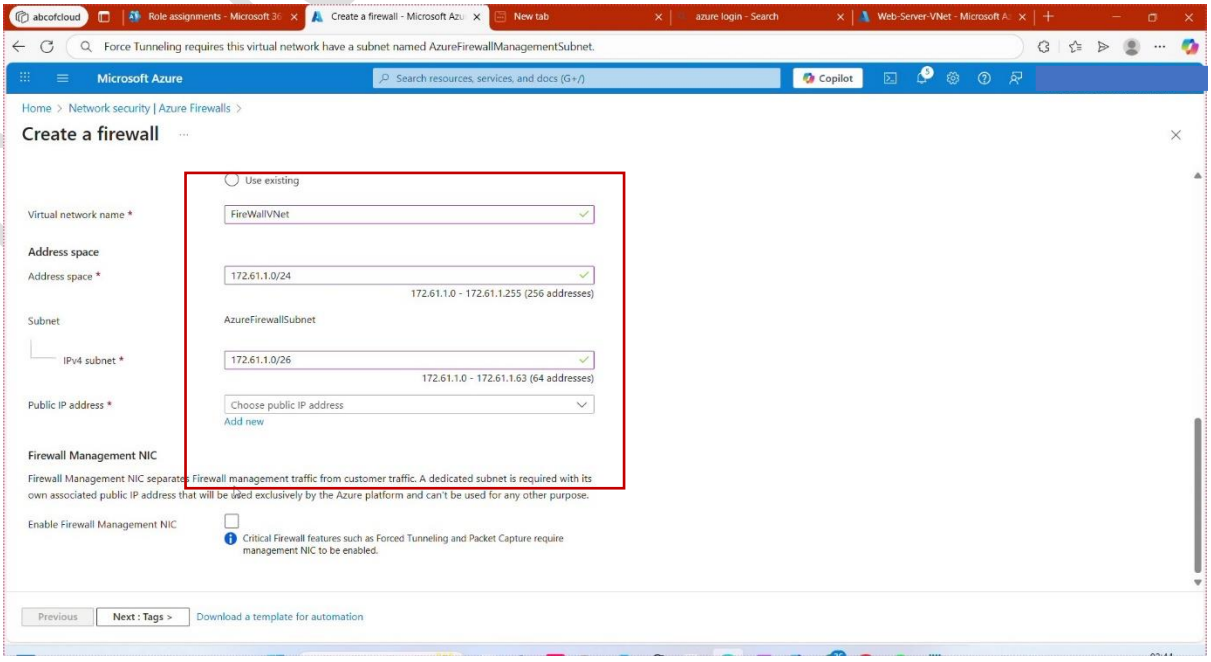
2. **Select Create Firewall.**

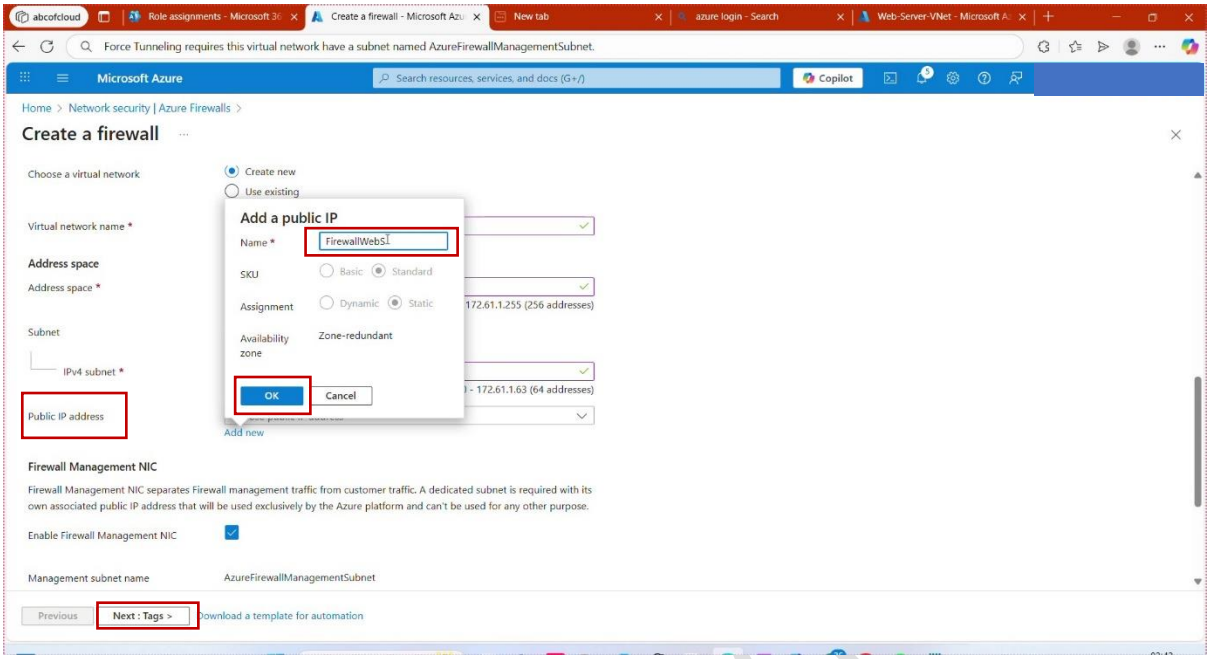


3. **Fill in the necessary fields**

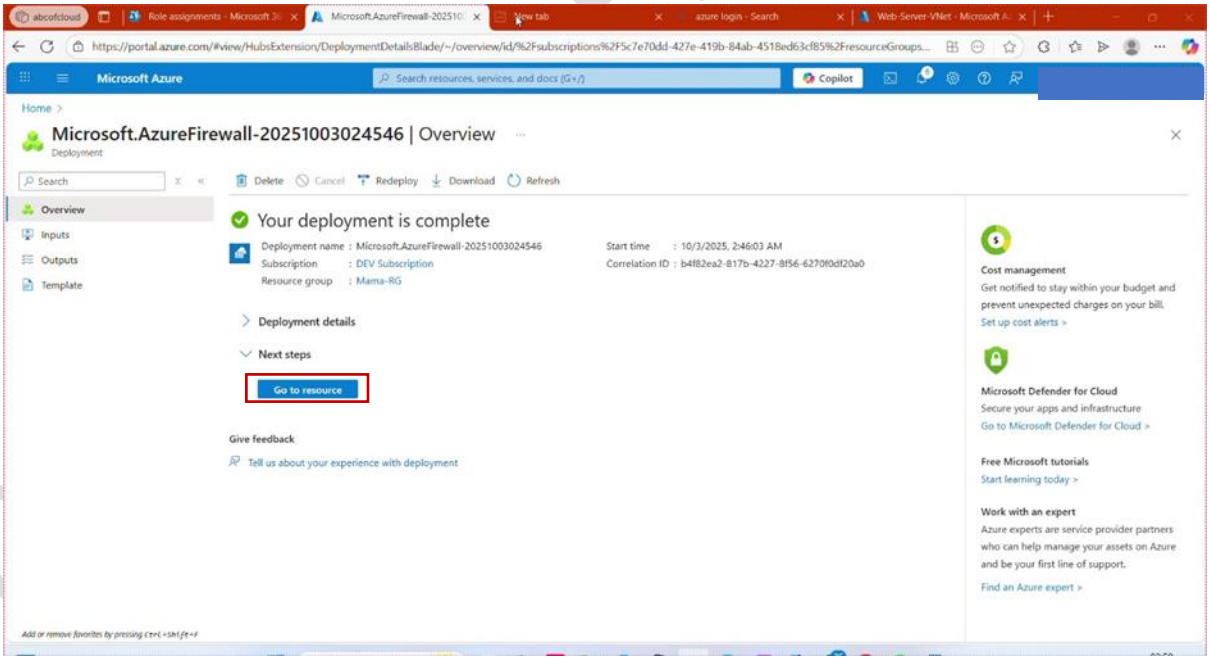
- Select your **subscription.**
- Select the **resource group** from the dropdown
- Give the firewall instance a **name**
- Select the same **location** with RG.
- Select **Standard** from the Firewall SKU selection boxes.
- Select Use **Firewall rules (classic)** to manage this firewall.
- Select Use **existing** or create a **new** VNet for the Firewall
- For **Public IP address**, select **Add new** and give it a **name** select **OK.**





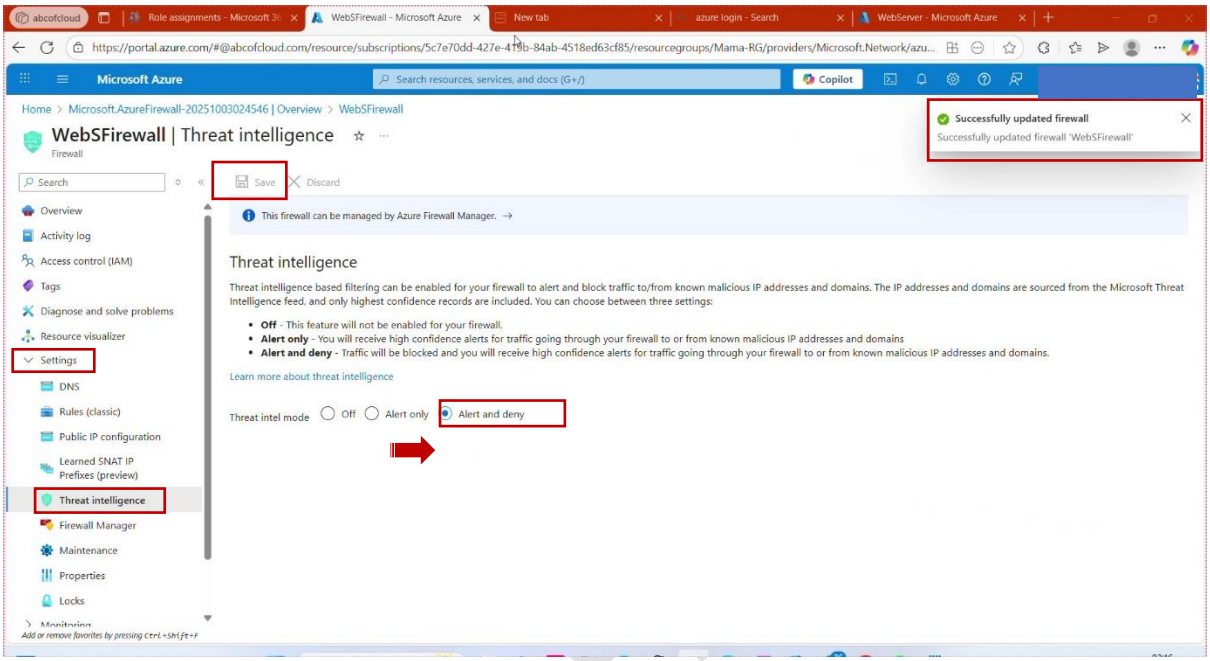


4. Select Review + create> create.

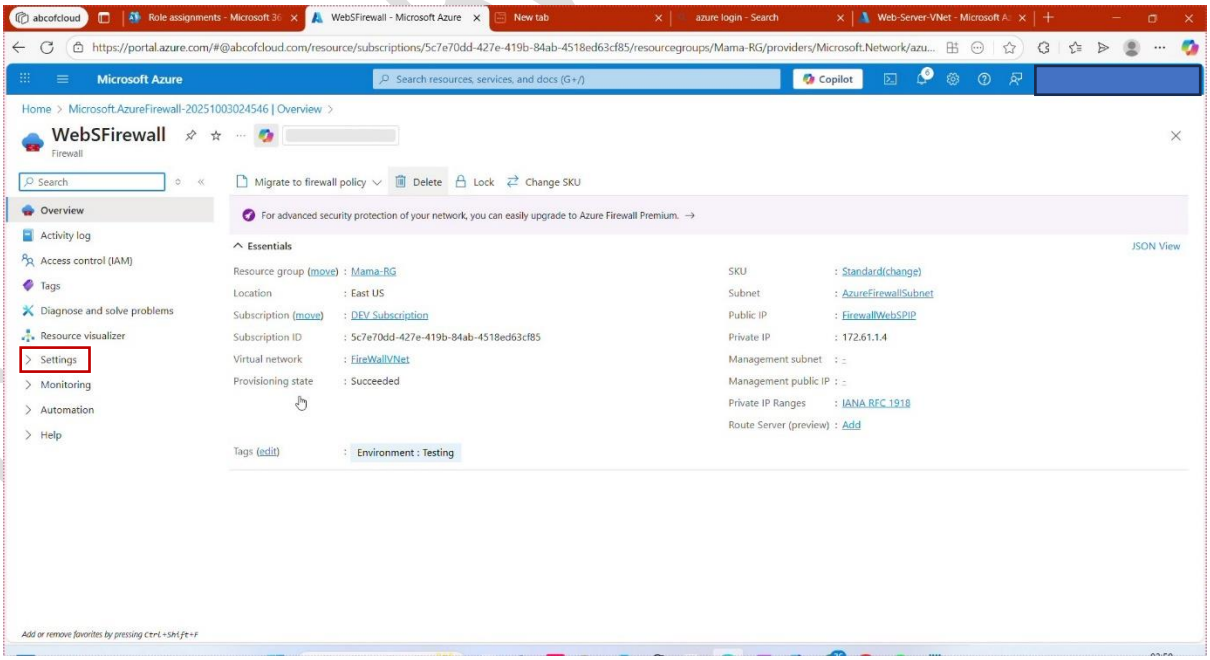


5. 0pen firewall resource, under **Settings**, select **Threat intelligence**.

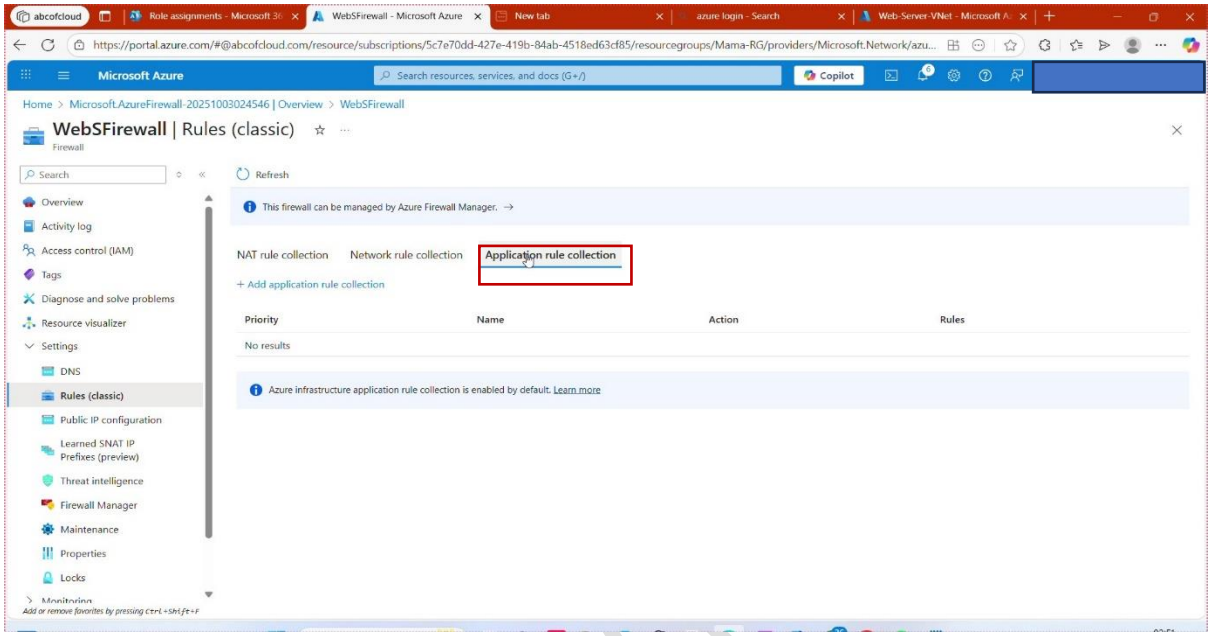
   >Threat intel mode: select **Alert and deny** > Select **Save.**



6. 0pen firewall resource, click on **Setting** select **Rules (classic).**

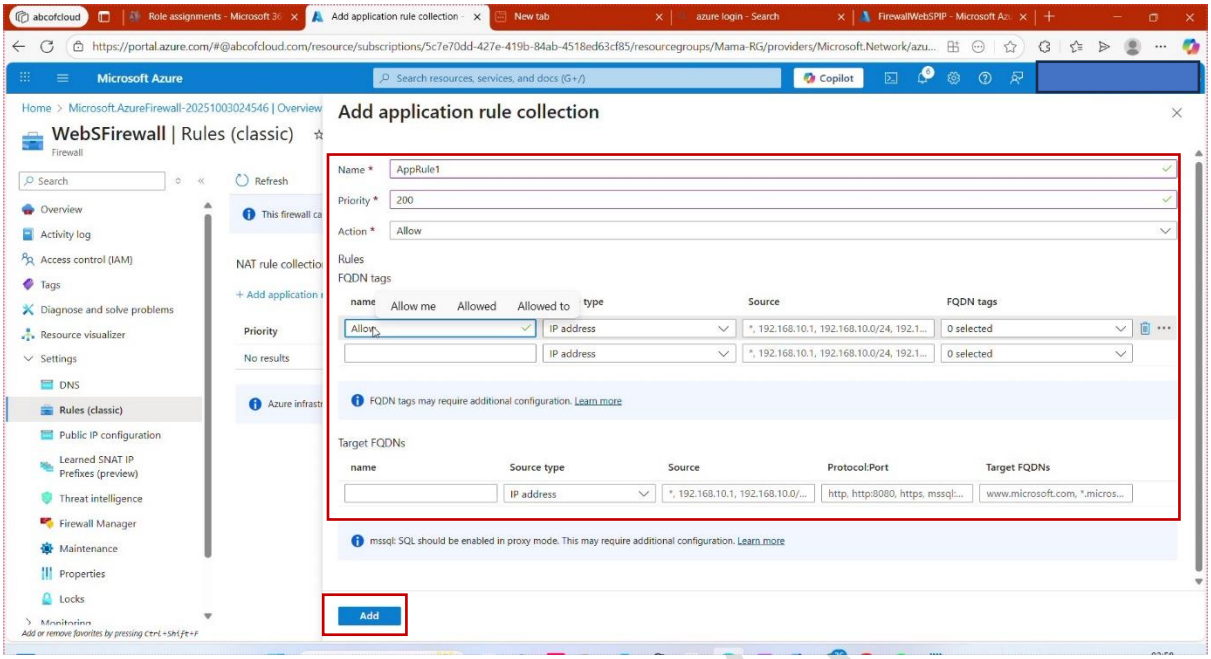


7. Select **Application rule** collection tab. **> Add application rule collection.**

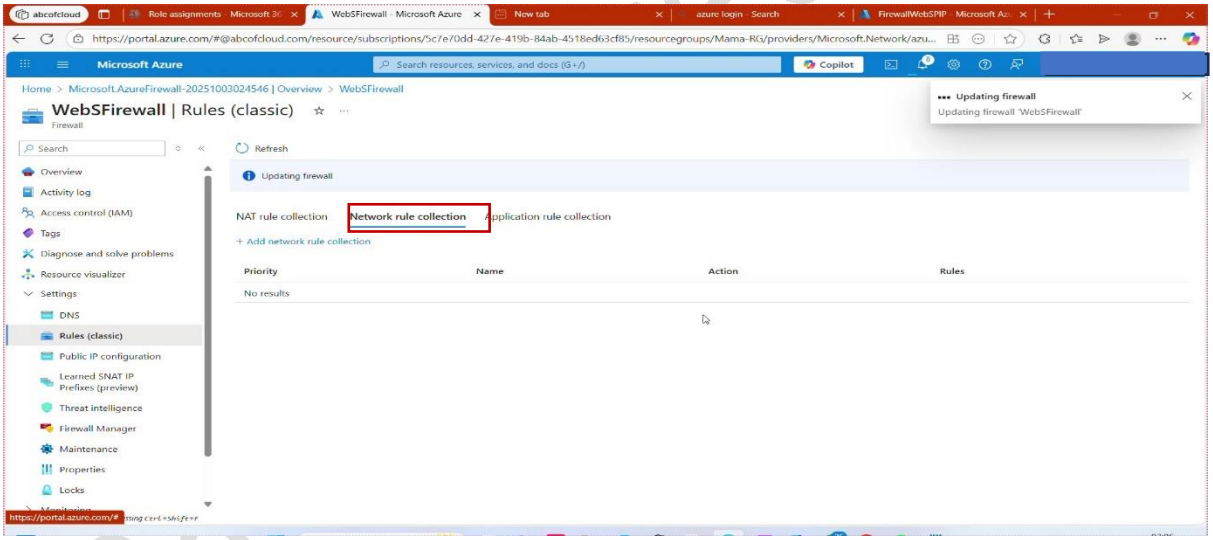


- For **Name,** type "AppRule1".
- For **Priority,** type "200".
- For **Action,** select **Allow.**
- Under **Rules, Target FQDNs,** for Name, type "Allow-Google".
- For **Source** type, select IP address.
- Type an IP address for the source.
- For **Protocol:port,** type "http, https".
- For **Target FQDNS**, type "www.google.com".
- Select **Add**

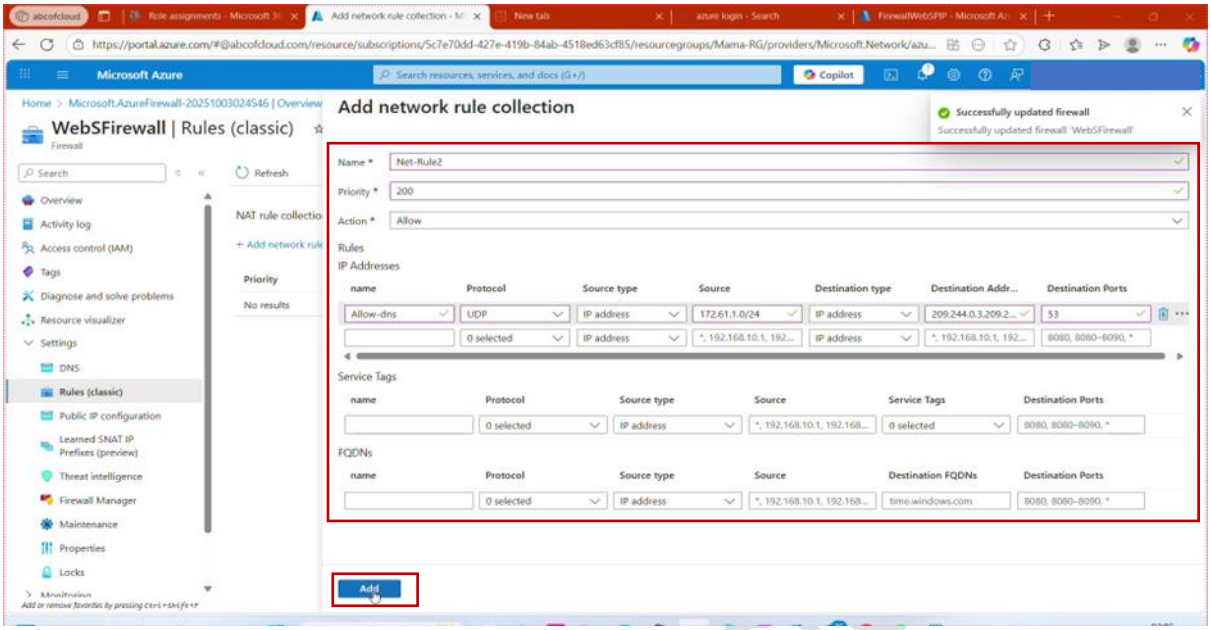
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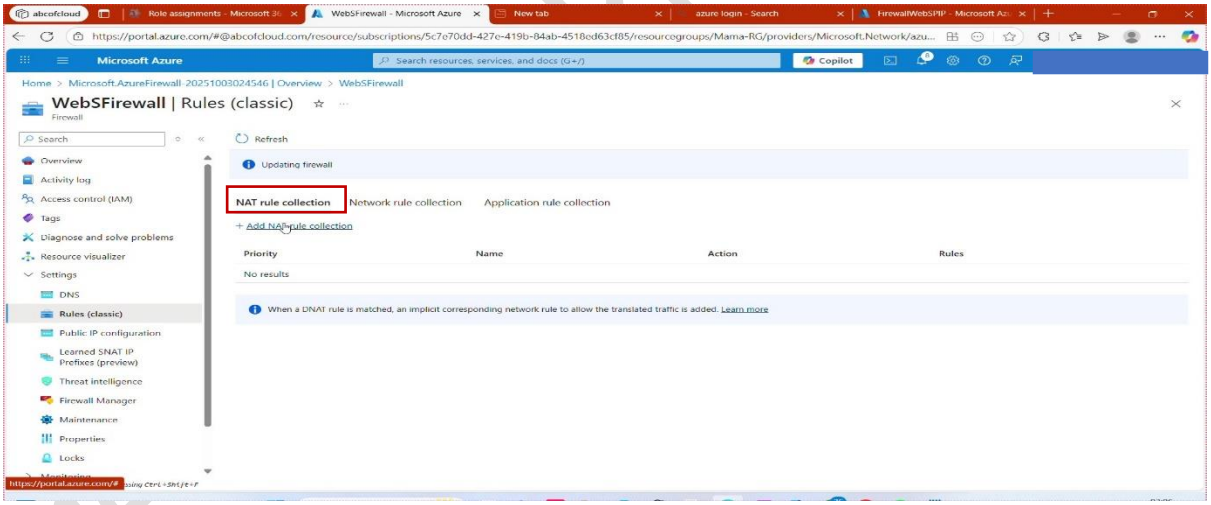
8. Select the **Network rule collection** tab. > **Add network rule collection.**



- For **Name, type** "Net-Rule1"
- For **Priority, type** "200".
- For **Action**, select Allow.
- Under **Rules**, IP addresses, for Name, type "Allow-DNS".
- For **Protocol**, select UDP.
- For **Source** type, select IP address.
- Type your Ip address for the **Source.**
- For **Destination type** select IP address.
- For **Destination address,** type (Public DNS server IP addresses 209.244.0.3,209.244.0.4
- For **Destination Ports**, type "53".
- Select **Add**

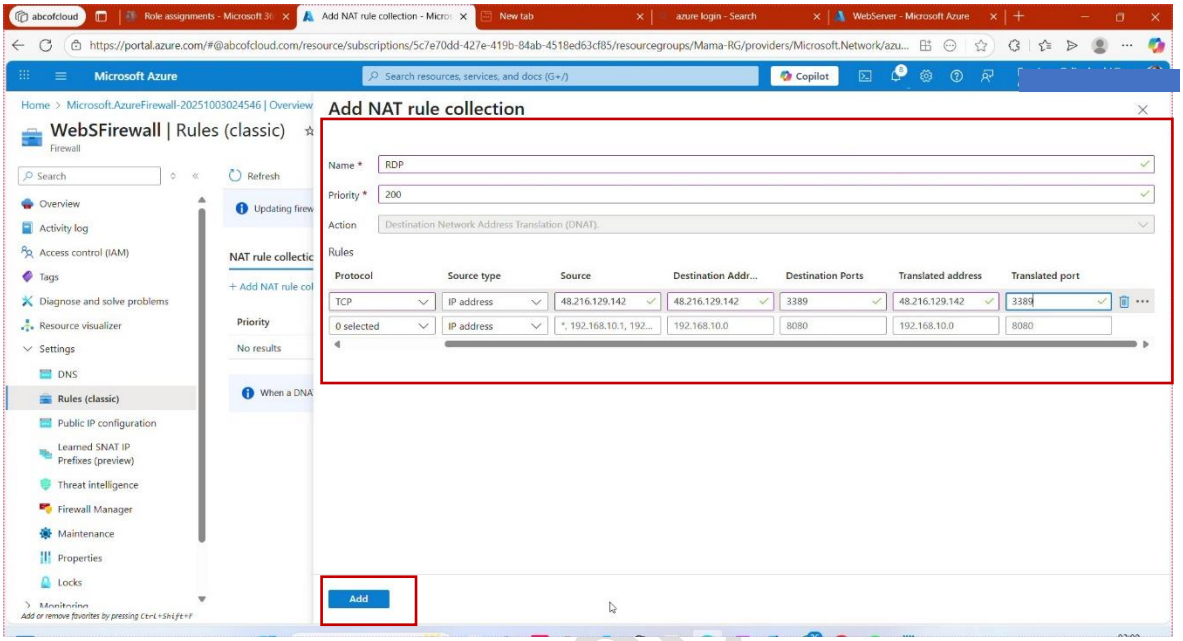


9. Select the **NAT rule collection > Add NAT rule collection.**



- For **Name,** type " rdp".
- For **Priority,** type "200".
- Under **Rules,** for Name, type "rdp-nat2".
- For **Protocol,** select TCP.
- For **Source type,** select IP address.
- Type (anything) for **the Source.**
- For **Destination** address, type the firewall public IP address.
- For **Destination Ports,** type "3389".
- For **Translated address,** type your virtual machines private IP address.

- For **Translated port,** type "3389". > Select **Add**



Conclusion

This project showcases a complete Azure Firewall deployment with layered rule sets and threat protection. It reflects best practices in cloud security architecture and demonstrates how to build a scalable and secure perimeter in Microsoft Azure.

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